Parsing – PDAs
Compilers

Consider the following grammar:

$$\begin{array}{lcl} C & ::= & L \mid L \vee C \\ L & ::= & a \mid \neg a \end{array}$$

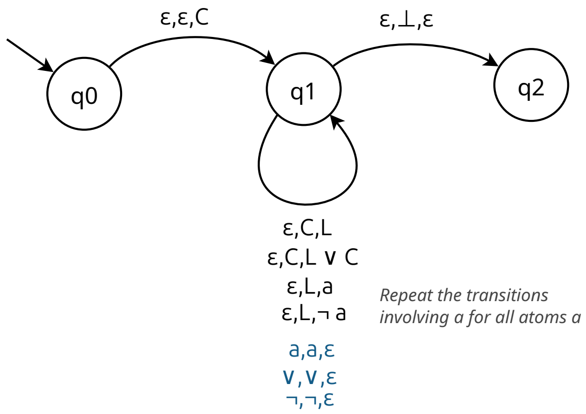
This defines Conjunctive Normal Form (CNF) clauses, which are disjunctions ($\vee$s) of literals, i.e., atoms a, or negations of atoms of the form $\neg\, a$. We write here a for any atom a, b, c, etc. For example, the following formulas are CNF clauses:

- a
- $\neg\, a$
- $a \vee b$
- $a \vee \neg\, b \vee c$

1. Write a top-down parser for this grammar as a PDA.
2. Show that $a \vee \neg\, b$ is accepted by your PDA.
3. Show that $a \vee\vee\, b$ is rejected by your PDA.
4. Write a bottom-up parser for this grammar as a PDA.
5. Show that $\neg\, a \vee b$ is accepted by your PDA.

Solution

1. Here is a top-down parser for this grammar, where the initial stack symbol is $\perp$ and where the blue transitions consume inputs:



2. Let us show that $a \vee \neg\ b$ is accepted by this PDA:

   - $(q_0, a \vee \neg\ b, \perp)$
   - $\rightarrow (q_1, a \vee \neg\ b, C\ \perp)$
   - $\rightarrow (q_1, a \vee \neg\ b, L \vee C\ \perp)$
   - $\rightarrow (q_1, a \vee \neg\ b, a \vee C\ \perp)$
   - $\rightarrow (q_1, \vee \neg\ b, \vee\ C\ \perp)$
   - $\rightarrow (q_1, \neg\ b, C\ \perp)$
   - $\rightarrow (q_1, \neg\ b, L\ \perp)$
   - $\rightarrow (q_1, \neg\ b, \neg b\ \perp)$
   - $\rightarrow (q_1, b, b\ \perp)$
   - $\rightarrow (q_1, \epsilon, \perp)$
   - $\rightarrow (q_2, \epsilon, \epsilon)$

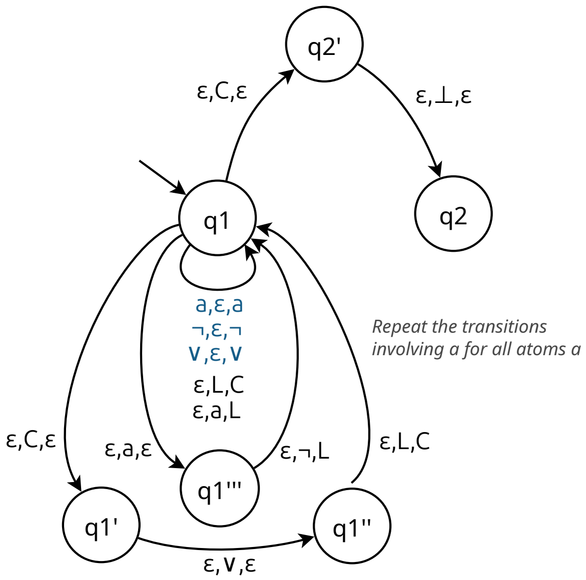
   Therefore $a \vee \neg b$ is accepted.

3. Let us show that $a \vee\vee\ b$ is accepted by this PDA:

   - $(q_0, a \vee\vee\ b, \perp)$
   - $\rightarrow (q_1, a \vee\vee\ b, C\ \perp)$
   - $\rightarrow (q_1, a \vee\vee\ b, L \vee C\ \perp)$
   - $\rightarrow (q_1, a \vee\vee\ b, a \vee C\ \perp)$
   - $\rightarrow (q_1, \vee\vee\ b, \vee\ C\ \perp)$
   - $\rightarrow (q_1, \vee\ b, C\ \perp)$

   The reduction will be stuck whatever we expand C to.

4. Here is a bottom-up parser for this grammar, where the initial stack symbol is $\perp$ and where the blue transitions indicate the shifts:



5. Let us show that $\neg a \vee b$ is accepted by this PDA:

   - $(q_1, \neg a \vee b, \perp)$
   - $\to (q_1, a \vee b, \neg\perp)$
   - $\to (q_1, \vee b, a \neg\perp)$
   - $\to (q_1''', \vee b, \neg\perp)$
   - $\to (q_1, \vee b, L\perp)$
   - $\to (q_1, b, \vee L\perp)$
   - $\to (q_1, \epsilon, b \vee L\perp)$
   - $\to (q_1, \epsilon, L \vee L\perp)$
   - $\to (q_1, \epsilon, C \vee L\perp)$
   - $\to (q_1', \epsilon, \vee L\perp)$
   - $\to (q_1'', \epsilon, L\perp)$
   - $\to (q_1, \epsilon, C\perp)$
   - $\to (q_2', \epsilon, \perp)$
   - $\to (q_2, \epsilon, \epsilon)$

   Therefore $\neg a \vee b$ is accepted.